# An opinionated comparison of model-based and classical ordination methods

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# Outline

- Part I: Overview
  - Model-based ordination
  - Compared to classical methods
- Part II: Examples
  - Cover classes
  - Biomass
  - Continuous cover with complete presence/absence
- Part III: Summary
  - Contrast to classical methods

# Limitations of classical multivariate analysis

Classical ordination methods are limited:

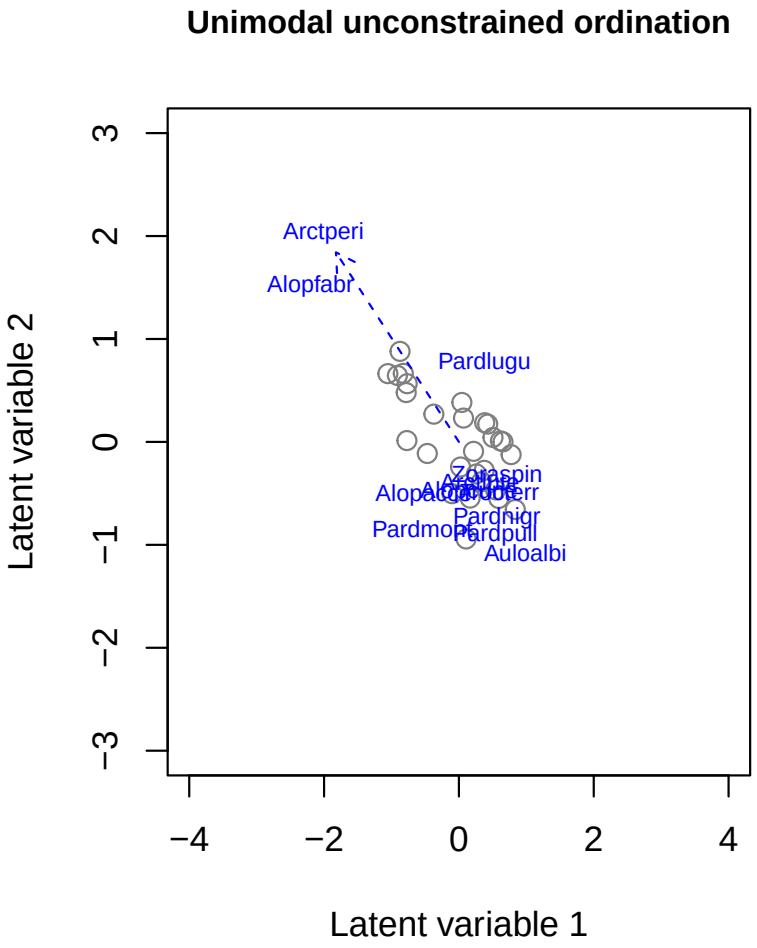
- Loose framework of separate methods
- No explicit model structure
- Cannot accommodate data properties
- Strong assumptions about data properties and structure
- Assumptions are not verifiable
- No uncertainty
- Lacks predictive capacity

# Model-based ordination is the next step

Classical ordination stops at the biplot. NMDS and distance-based methods traded process for pattern.

With model-based ordination, the biplot is only the beginning.

It is also where ordination was headed (ter Braak and Prentice 1988, ter Braak 1985, Ihm and van Groenewoud 1975, Gauch et al. 1974) prior to Minchin (1987) popularizing NMDS



# What is model-based ordination

Distribution

$$y_{ij} \sim f(\mu_{ij}, \phi_j)$$

# What is model-based ordination

Link function

$$y_{ij} \sim f(\mu_{ij}, \phi_j)$$

$$g(\mu_{ij}) = \beta_{0j} + \mathbf{u}_i^\top \boldsymbol{\gamma}_j$$

Ordination

Species intercepts

# What is model-based ordination

Site intercepts

Link function

$$y_{ij} \sim f(\mu_{ij}, \phi_j)$$

$$g(\mu_{ij}) = \alpha_i + \beta_{0j} + \mathbf{u}_i^\top \boldsymbol{\gamma}_j$$

$$\alpha_i = \mathbf{w}_i^\top \boldsymbol{\delta} + a_i, \quad \mathbf{a} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}^a), \quad \mathrm{vec}(\mathbf{B}) \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}^r \otimes \boldsymbol{\Sigma})$$

Ordination

Fixed effects

Random effect

Species intercepts

# What is model-based ordination

Site intercepts

Environmental predictors (with species coefficients)

Link function

$$y_{ij} \sim f(\mu_{ij}, \phi_j)$$

$$g(\mu_{ij}) = \alpha_i + \beta_{0j} + \mathbf{x}_i^\top \boldsymbol{\beta}_j + \mathbf{z}_i^\top \mathbf{b}_j + \mathbf{u}_i^\top \boldsymbol{\gamma}_j$$

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Ordination

Species random effects

Fixed effects

Random effect

Species intercepts

# Key advantages of Model-based ordination

- **Distribution-based**: no distances or data transformations; direct connection to data is maintained
- **Random effects**: integrates ordination with mixed-effects modeling for hierarchical designs
- **Diagnostics and inference**: enables assumption checking, model evaluation, uncertainty quantification
- **Predictive capacity**: enables (multivariate) forecasting
- **Diversity estimation**: facilitates inference on diversity in the same framework

# Part II: Examples

Each example steps further from what classical ordination can do:

1. Cover classes from Skabbholmen, Sweden
   - design-aware ordination: random effects change interpretation
2. Biomass data from the Waddensea, Germany
   - ordination and prediction are the same model
3. Cover data from Rothamsted, United Kingdom
   - predicting an emergent community property: species richness

# Example 1: Skabbholmen, Sweden

- **Sites**: 64 unique plots across 4 transects
- **Species**: 49 vascular plant species with cover data on a 5-degree Hult-Sernander-Du Rietz scale
- **Predictors**:
  - Year (1978 vs 1984) - representing temporal change
  - Elevation (centimeters) - representing long-term land uplift
- **Data type**: Cover classes
- **Analysis**: Cumulative probit/logit model with 2 informed LVs

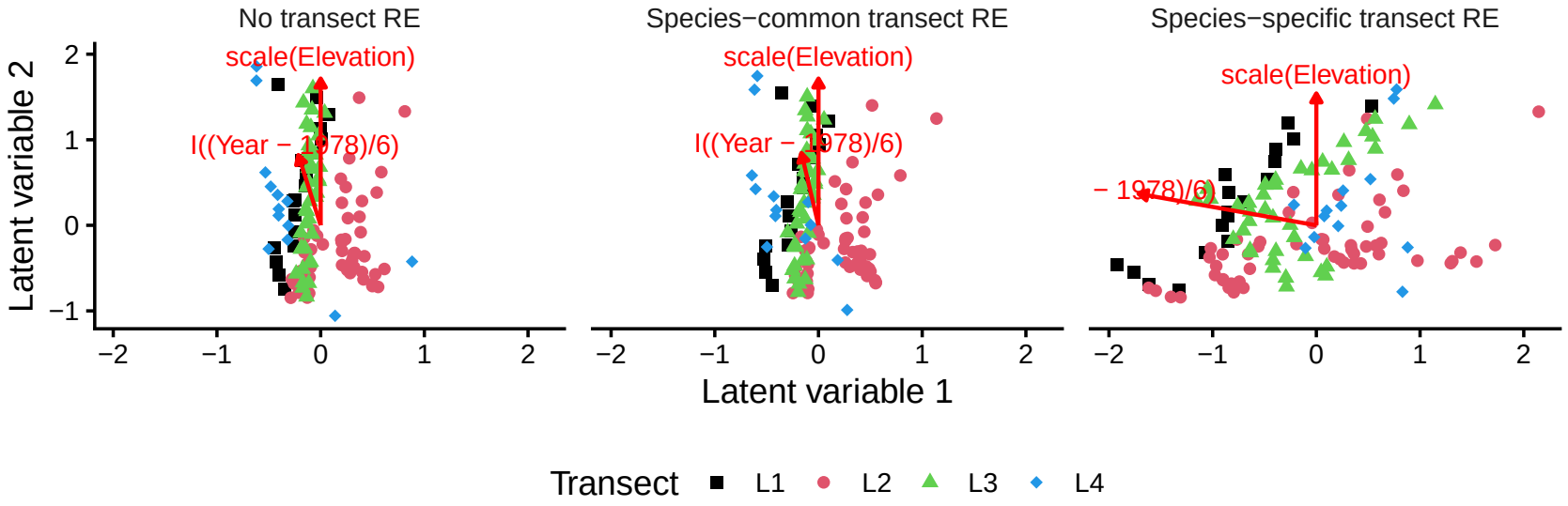
Previously analyzed by ter Braak (1987)
Originally by Cramer and Hytteborn (1987)

# Example 1: Does design matter for ordination?

Three versions of the same model — differing only in how transect structure is handled:

- **No transect RE**: latent variables must absorb any spatial clustering between transects
- **Species-common transect RE**: one intercept shift per transect, shared across all species
- **Species-specific transect RE**: each species gets its own transect random effect

# Example 1: Random effects change the ordination




- Without a transect RE, transect effect are confounded with the environmental space: the axes reflect clustering
- Adding the RE removes the confounding, the latent variables align with the elevation and year effects
- The choice of RE structure is a modelling decision — one that classical ordination cannot make

# Example 2: Wadden Sea, Germany

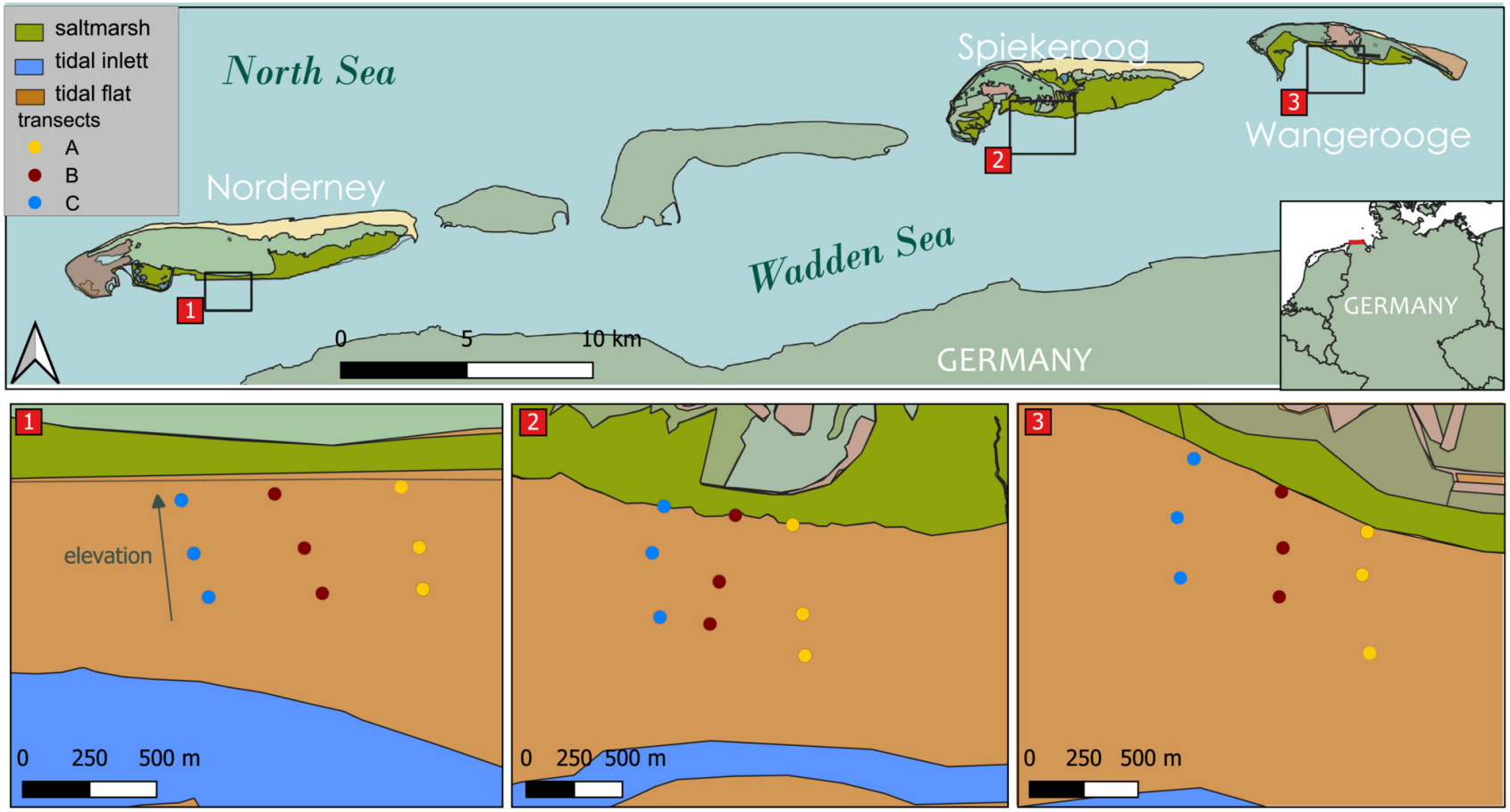


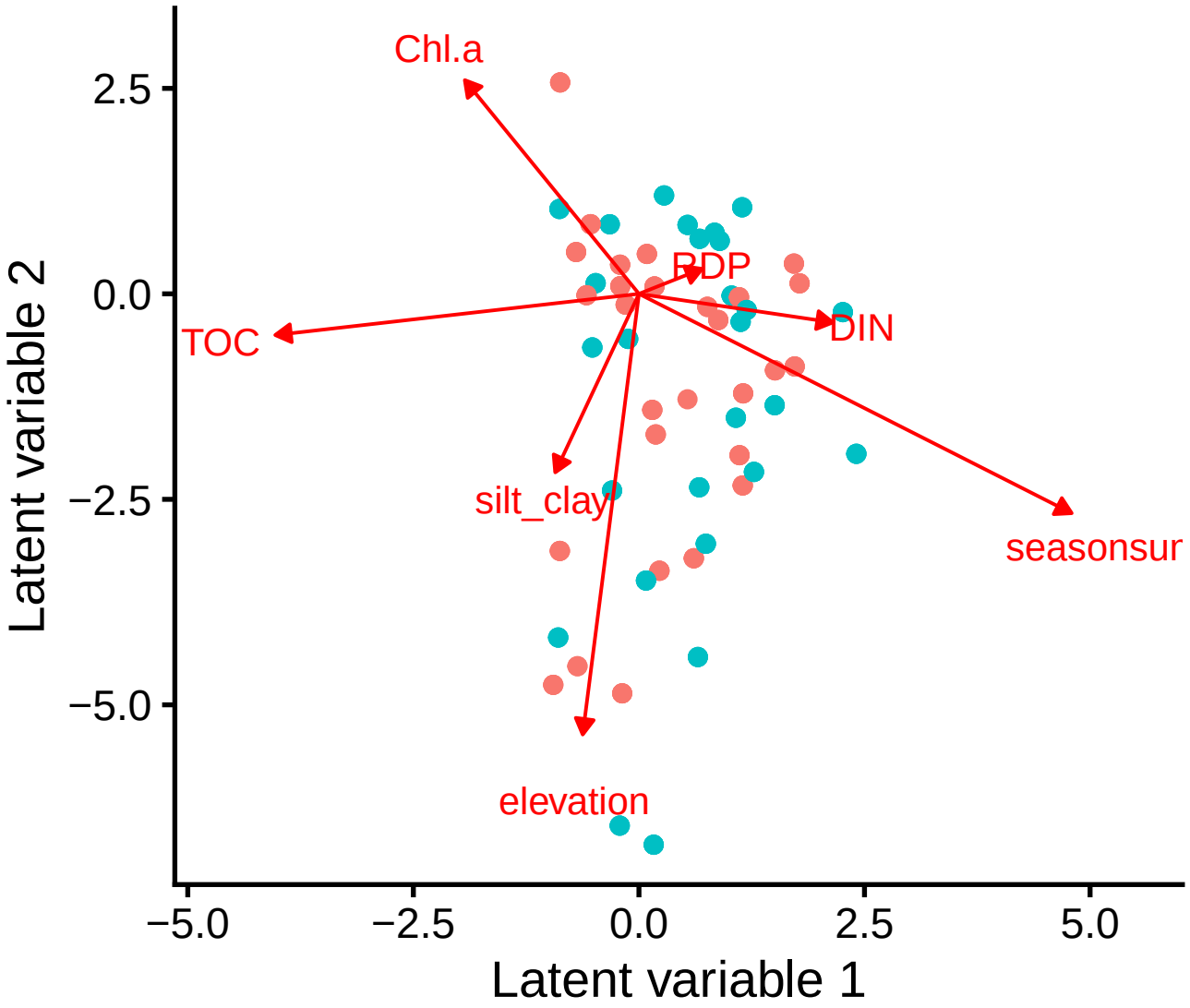

Figure 1: nioz.nl

# Example 2: Wadden Sea, Germany

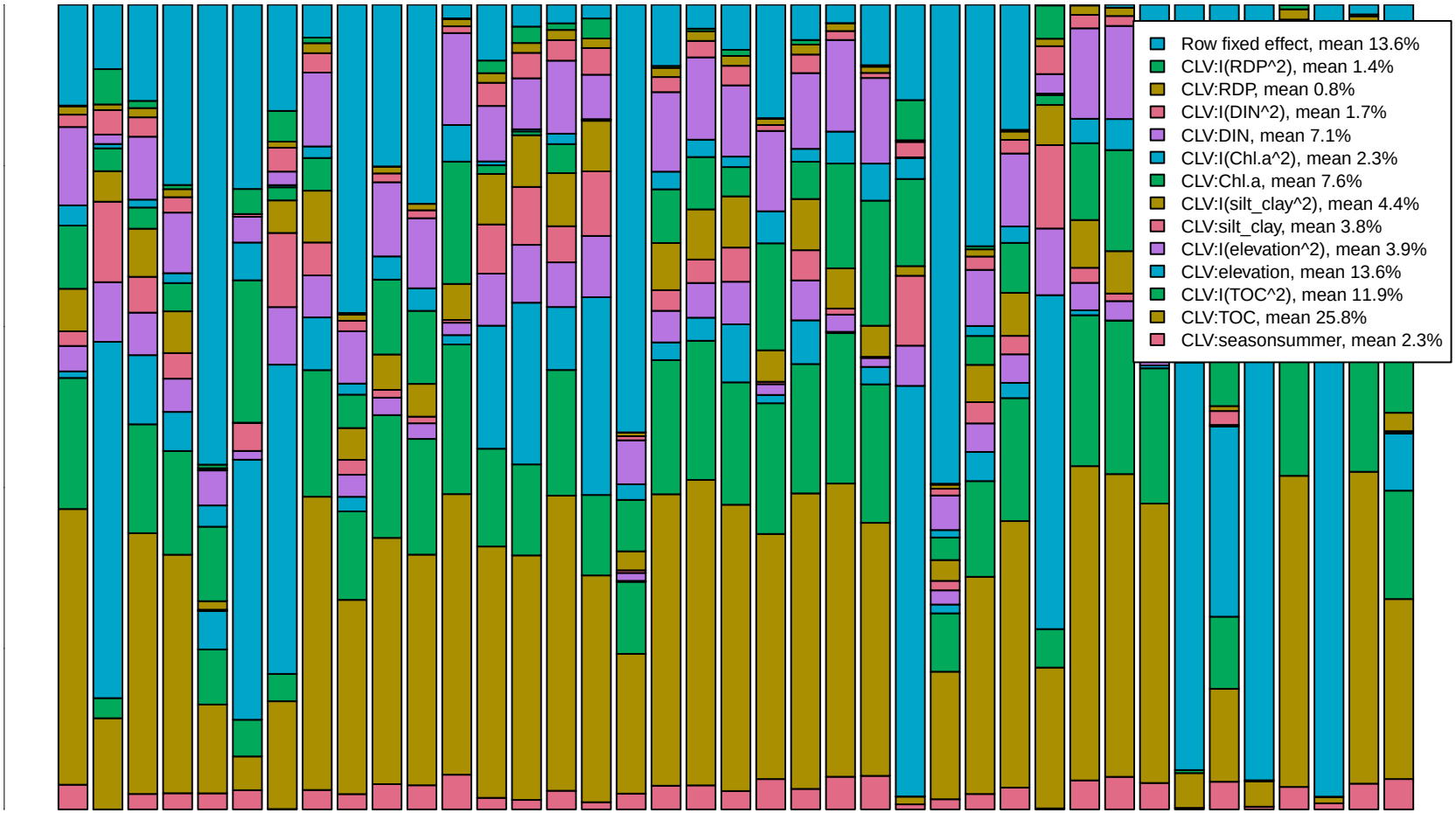
- **Sites**: 3 sites at 3 transects on each of 3 islands
- **Species**: 58 macrozoobenthos species
- **Predictors**:
  - Season - representing temporal variation
  - TOC - total organic carbon in sediment
  - Elevation - elevation of sediment surface
  - Mud - percentage of mud in sediment
  - Chlorophyll a - primary production
  - DIN - dissolved inorganic nitrogen
  - RDP - respiration rate of sediment
- **Data type**: Biomass data
- **Analysis**: Tweedie log-linear model with 2 constrained latent variables

Originally by Dewenter et al. (2023)

# Example 2: Constrained ordination

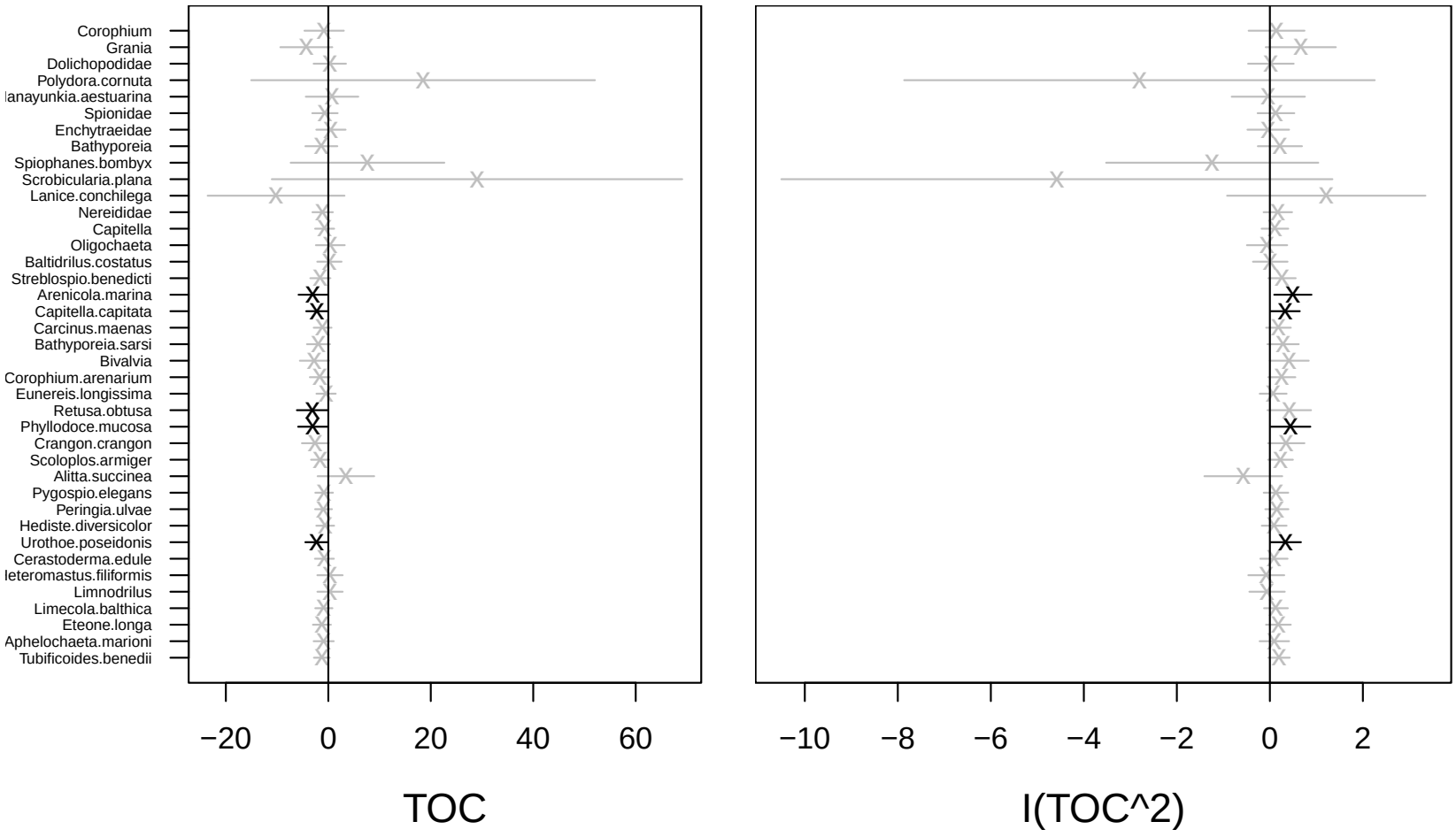


# Example 2: Variance partitioning



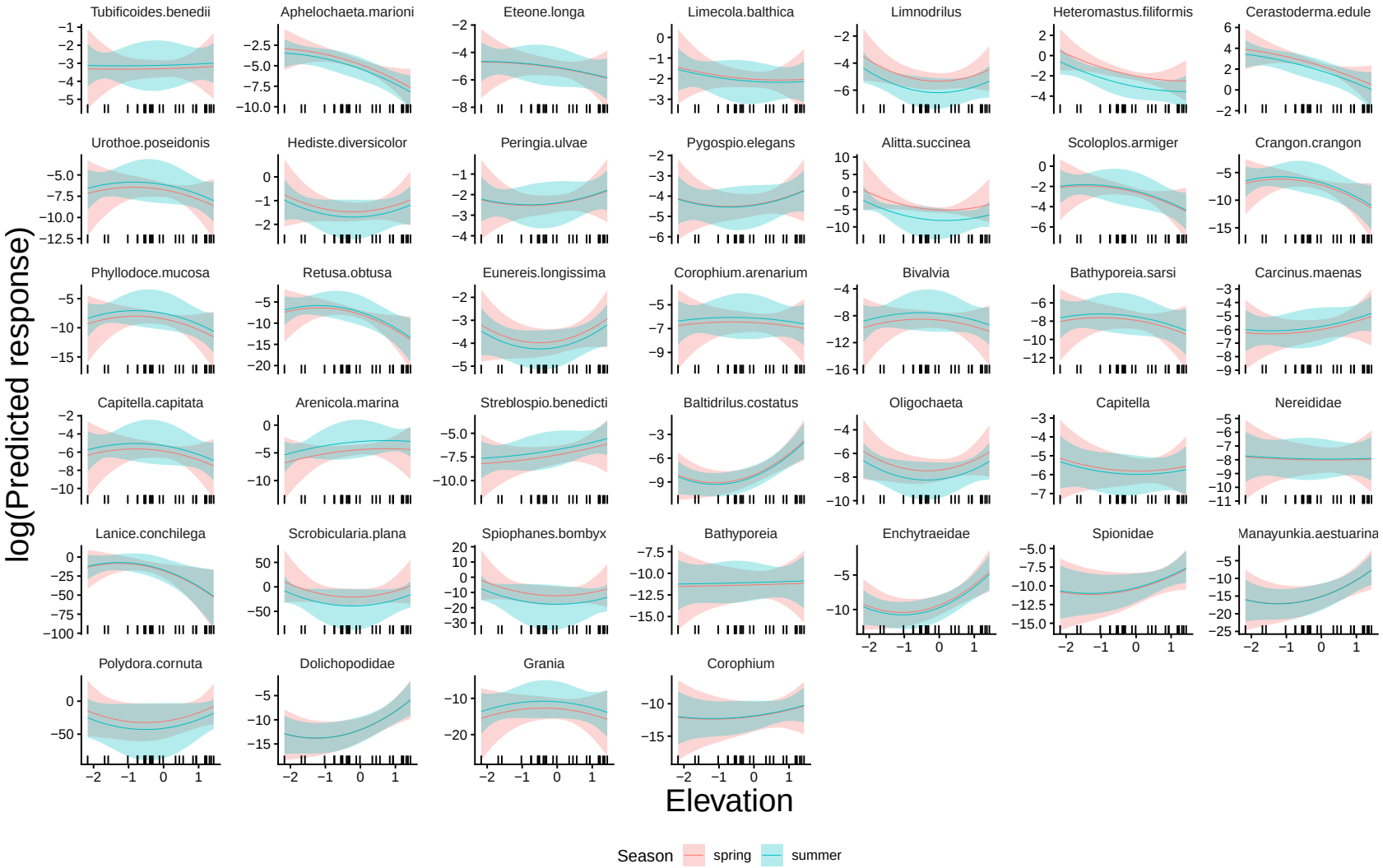

Variance partitioning attributes a large share of explained variance to TOC — suggesting it is an important driver.

# Example 2: Species-specific results




Yet species-specific TOC effects are highly uncertain and vary widely across species — variance alone is not enough.

# Example 2: Species-level predictions



# Example 3: Rothamsted, United Kingdom

- **Sites**: 16 plots, each with subplots
- **Species**: 57 vascular plant species
- **Predictors**:
  - Fertilizer treatment
    - Phosphorus, Potassium, Nitrogen, Liming, and so on
- **Data type**: Continuous cover with absence and presence
- **Analysis**: Ordered beta model with two constrained latent variables

Originally studied by a.o. Crawley et al. (2005)

# Example 3: From ordination to diversity

Species richness is not modelled directly — it emerges from the joint species model:

- No separate richness model is needed — richness emerges from the joint species model for any data type
- Uncertainty propagates naturally from species-level predictions, avoiding distributional assumptions on richness directly

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Both the species responses and the community-level summary are coherently estimated in one step.

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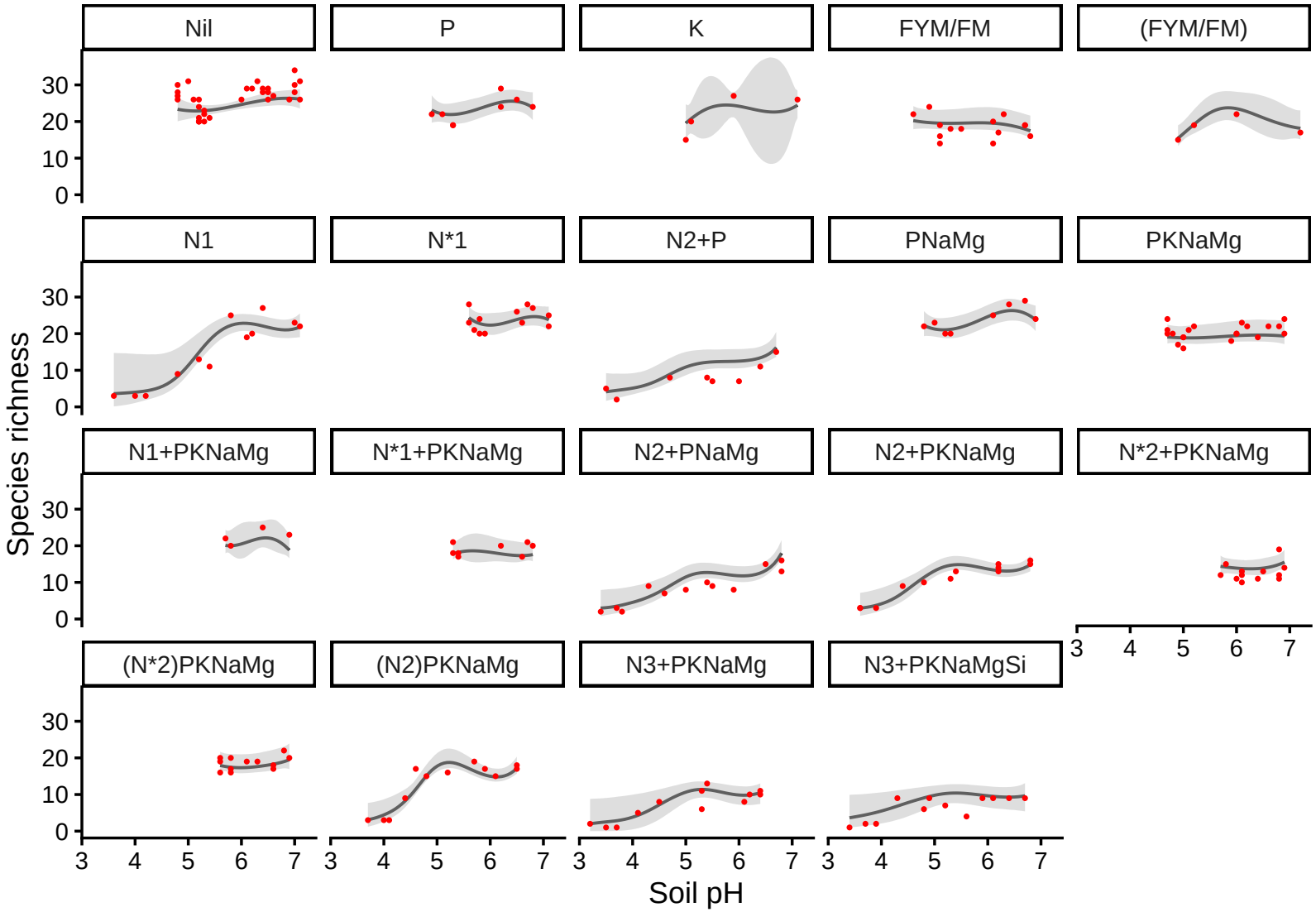

# What the examples show

## and what classical ordination cannot do

1. **Skabbholmen**: Ignoring hierarchical design confounds the ordination
2. **Wadden Sea**: With model-based ordination species-level forecasting comes for free
3. **Park Grass**: When the species data is modelled, community-level metrics follow

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1. No hierarchical design: RDA/CCA cannot incorporate random effects
2. Stop at the biplot: there is no prediction or uncertainty
3. Severs the data connection: you cannot extract diversity

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Pattern visualization → design-aware inference → multivariate prediction → community-level forecasting.

# Summary

Model-based ordination is not more complex, but more complete:

- All data types handled in a single, coherent framework
- Non-linearity: what NMDS approximates via ranks, model-based ordination handles explicitly via the link function and model
- Rare species: data deficiency offset by partial pooling across species
- Assumptions are explicit, verifiable, and can be relaxed
- Uncertainty is propagated, not ignored
- Hierarchical designs are accommodated naturally
- Prediction is built in, not bolted on

# Future directions

- Diversity measures directly from multispecies models (van der Veen in prep., Many authors et al. in prep)
- Temporal; Rugstad et al. in prep.
- Spatial; Korhonen et al. in prep
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- Structured species' random effects
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I haven't really spoken about species' correlation in a JSDM context, but it's here.